DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-100-01 — Internal inspection of tank T-101

| | | | |
|---|---|---|---|
| **Area** | A100 Tank farm | **Task** | TI Tank internal inspection |
| **Priority** | 2 (integrity) | **Issued by** | Facility owner (demonstration) |
| **Primary asset** | T-101 — T-101 tank 12' dia × 22' high — E 25'-0"  N 26'-6"  EL 100'-6" | | |
| **Equipment** | T-101 | **Lines** | 6"-P150-P-1001, 4"-P150-P-6038, 1"-P150-V-6030, 1"-P150-D-6031, 1"-P150-N-3015, 2"-P300-LS-3007, 2"-P150-SC-3008, 3/4"-P150-I-6032, 3/4"-P150-I-6033 |
| **Isometrics** | ISO-6"-P150-P-1001, ISO-4"-P150-P-6038, ISO-1"-P150-V-6030, ISO-1"-P150-D-6031, ISO-1"-P150-N-3015, ISO-2"-P300-LS-3007, ISO-2"-P150-SC-3008, ISO-3/4"-P150-I-6032, ISO-3/4"-P150-I-6033 (generated); owner's drawings ISO-P-1001, ISO-P-6038, ISO-V-6030, ISO-D-6031, ISO-N-3015, ISO-LS-3007, ISO-SC-3008, ISO-I-6032, ISO-I-6033 | | |
| **Permits** | General work permit, Confined-space entry permit | | |

## Reason

Scheduled internal inspection of the shell, floor and roof.

## Scope of work

1. Isolate T-101 per the isolation list; drain, vent, purge and gas-test for entry.
2. Open for entry; clean as needed for inspection.
3. Internal visual examination; ultrasonic thickness at the 15 TMLs; floor scan.
4. Close with new gaskets on every opened joint; remove blinds; restore the line-up.

Affected items: T-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

## Owner's isometric drawings (SW-IMP-6)

| Drawing | Line drawn | Work line | Sheets | Match to the plant model |
|---|---|---|---|---|
| ISO-P-1001 | 6"-P150-P-1001 | 6"-P150-P-1001 | 5 | 1 error(s), 4 warning(s) — see below |
| ISO-P-6038 | 4"-P150-P-6038 | 4"-P150-P-6038 | 1 | agrees with the model |
| ISO-V-6030 | 1"-P150-V-6030 | 1"-P150-V-6030 | 1 | agrees with the model |
| ISO-D-6031 | 1"-P150-D-6031 | 1"-P150-D-6031 | 1 | agrees with the model |
| ISO-N-3015 | 1"-P150-N-3015 | 1"-P150-N-3015 | 1 | 0 error(s), 1 warning(s) — see below |
| ISO-LS-3007 | 2"-P300-LS-3007 | 2"-P300-LS-3007 | 1 | agrees with the model |
| ISO-SC-3008 | 2"-P150-SC-3008 | 2"-P150-SC-3008 | 1 | 0 error(s), 2 warning(s) — see below |
| ISO-I-6032 | 3/4"-P150-I-6032 | 3/4"-P150-I-6032 | 1 | agrees with the model |
| ISO-I-6033 | 3/4"-P150-I-6033 | 3/4"-P150-I-6033 | 1 | agrees with the model |

ISO-P-1001 (PDF): ../isometrics-imported/ISO-P-1001.pdf · ISO-P-6038 (PDF): ../isometrics-imported/ISO-P-6038.pdf · ISO-V-6030 (PDF): ../isometrics-imported/ISO-V-6030.pdf · ISO-D-6031 (PDF): ../isometrics-imported/ISO-D-6031.pdf · ISO-N-3015 (PDF): ../isometrics-imported/ISO-N-3015.pdf · ISO-LS-3007 (PDF): ../isometrics-imported/ISO-LS-3007.pdf · ISO-SC-3008 (PDF): ../isometrics-imported/ISO-SC-3008.pdf · ISO-I-6032 (PDF): ../isometrics-imported/ISO-I-6032.pdf · ISO-I-6033 (PDF): ../isometrics-imported/ISO-I-6033.pdf — the generated isometrics follow the ITP.

| Drawing | Sheet | Level | Finding |
|---|---|---|---|
| ISO-P-1001 | 5 | error | sheet 5: CONT'D ON ISO-P-1002 prints E 154'-5" N -12'-6" EL 102'-0" for E 154'-5" N -11'-6" EL 102'-0": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 6"-P150-P-1001 meets 6"-P150-P-1002 (E 154'-5" N -11'-6" EL 102'-0") |
| ISO-P-1001 | 5 | warning | W21 (6"-P150-P-1001/W21, sheet 5) is labelled again on sheet 4; it is 58.0 in from the nearest break |
| ISO-P-1001 | 4 | warning | branch weld 3/4"-P300-I-7030/W1 (E 151'-6" N 0" EL 117'-5", sheet 4) has no label |
| ISO-P-1001 | 4 | warning | branch weld 3/4"-P300-I-7001/W1 (E 152'-6" N 0" EL 117'-5", sheet 4) has no label |
| ISO-P-1001 | 5 | warning | TO label 'P-101A NOZ. S' is not on the drawing |
| ISO-N-3015 | 1 | warning | weld 1"-P150-N-3015/W1 (E 28'-0" N 3'-6" EL 122'-8", sheet 1) has no label |
| ISO-SC-3008 | 1 | warning | weld 2"-P150-SC-3008/W11 (E 23'-6" N -1'-8 11/16" EL 125'-0", sheet 1) has no label |
| ISO-SC-3008 | 1 | warning | weld 2"-P150-SC-3008/W1 (E 23'-6" N -2'-0" EL 122'-8 1/2", sheet 1) has no label |

## Assets in the plant model (SW-WO-7)

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| T-101 | equipment | T-101 tank 12' dia × 22' high | | E 25'-0"  N 26'-6"  EL 100'-6" |
| 6"-P150-P-1001/A2 | item | VALVE, GATE, WEDGE, OS&Y 6" P150 | 6"-P150-P-1001 | E 32'-2-15/16"  N 26'-6"  EL 101'-6" |
| T-101-N1 | nozzle | Nozzle T-101-N1 (outlet, 6" P150) on T-101 | | E 31'-7-13/16"  N 26'-6"  EL 101'-6" |

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| 4"-P150-P-6038/2 | item | VALVE, GATE, WEDGE, OS&Y 4" P150 | 4"-P150-P-6038 | E 25'-0"  N 22'-6"  EL 123'-10-13/16" |
| T-101-N2 | nozzle | Nozzle T-101-N2 (inlet, 4" P150) on T-101 | | E 25'-0"  N 22'-6"  EL 123'-3-1/16" |
| 3"-P150-RV-4002/2 | item | VALVE, PRESSURE SAFETY VALVE 3" P150 INLET × 4" P150 OUTLET | 3"-P150-RV-4002 | E 22'-0"  N 26'-6"  EL 124'-0" |
| 1"-P150-V-6030/2 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-V-6030 | E 25'-0"  N 26'-6"  EL 123'-10-1/2" |
| T-101-N4 | nozzle | Nozzle T-101-N4 (vent, 1" P150) on T-101 | | E 25'-0"  N 26'-6"  EL 123'-3-11/16" |
| 1"-P150-D-6031/2 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-D-6031 | E 18'-0"  N 26'-6"  EL 101'-0" |
| T-101-N5 | nozzle | Nozzle T-101-N5 (drain, 1" P150) on T-101 | | E 18'-6-7/8"  N 26'-6"  EL 101'-0" |
| 1"-P150-N-3015/B2 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-N-3015 | E 28'-0"  N 26'-6"  EL 123'-8-1/4" |
| T-101-N6 | nozzle | Nozzle T-101-N6 (nitrogen, 1" P150) on T-101 | | E 28'-0"  N 26'-6"  EL 123'-1-7/16" |
| 2"-P300-LS-3007/B2 | item | VALVE, GATE, WEDGE, OS&Y 2" P300 | 2"-P300-LS-3007 | E 18'-5/8"  N 26'-6"  EL 102'-6" |
| T-101-N7 | nozzle | Nozzle T-101-N7 (steam in, 2" P300) on T-101 | | E 18'-6-1/8"  N 26'-6"  EL 102'-6" |
| 2"-P150-SC-3008/A2 | item | VALVE, GATE, WEDGE, OS&Y 2" P150 | 2"-P150-SC-3008 | E 18'-5/8"  N 26'-6"  EL 104'-6" |
| T-101-N8 | nozzle | Nozzle T-101-N8 (condensate out, 2" P150) on T-101 | | E 18'-6-1/8"  N 26'-6"  EL 104'-6" |
| 3/4"-P150-I-6032/2 | item | VALVE, BALL, FLOATING 3/4" P150 | 3/4"-P150-I-6032 | E 18'-1/4"  N 26'-6"  EL 107'-0" |
| T-101-N9 | nozzle | Nozzle T-101-N9 (instrument, 0.75" P150) on T-101 | | E 18'-7"  N 26'-6"  EL 107'-0" |
| 3/4"-P150-I-6033/2 | item | VALVE, BALL, FLOATING 3/4" P150 | 3/4"-P150-I-6033 | E 18'-1/4"  N 26'-6"  EL 117'-0" |
| T-101-N10 | nozzle | Nozzle T-101-N10 (instrument, 0.75" P150) on T-101 | | E 18'-7"  N 26'-6"  EL 117'-0" |

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | valve | 6"-P150-P-1001/A2 | 6"-P150-P-1001 | Close and lock VALVE, GATE, WEDGE, OS&Y 6" P150 (6"-P150-P-1001/A2) on T-101-N1 | E 32'-2-15/16"  N 26'-6"  EL 101'-6" | | |
| 2 | blind | T-101-N1 | 6"-P150-P-1001 | Fit a blind between 6"-P150-P-1001/A2 and the nozzle flange T-101-N1 | E 31'-7-13/16"  N 26'-6"  EL 101'-6" | | |
| 3 | valve | 4"-P150-P-6038/2 | 4"-P150-P-6038 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-P-6038/2) on T-101-N2 | E 25'-0"  N 22'-6"  EL 123'-10-13/16" | | |
| 4 | blind | T-101-N2 | 4"-P150-P-6038 | Fit a blind between 4"-P150-P-6038/2 and the nozzle flange T-101-N2 | E 25'-0"  N 22'-6"  EL 123'-3-1/16" | | |
| 5 | removal | 3"-P150-RV-4002/2 | 3"-P150-RV-4002 | Remove the relief valve on T-101-N3 and blind the nozzle | | | |
| 6 | valve | 1"-P150-V-6030/2 | 1"-P150-V-6030 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6030/2) on T-101-N4 | E 25'-0"  N 26'-6"  EL 123'-10-1/2" | | |
| 7 | blind | T-101-N4 | 1"-P150-V-6030 | Fit a blind between 1"-P150-V-6030/2 and the nozzle flange T-101-N4 | E 25'-0"  N 26'-6"  EL 123'-3-11/16" | | |
| 8 | valve | 1"-P150-D-6031/2 | 1"-P150-D-6031 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6031/2) on T-101-N5 | E 18'-0"  N 26'-6"  EL 101'-0" | | |
| 9 | blind | T-101-N5 | 1"-P150-D-6031 | Fit a blind between 1"-P150-D-6031/2 and the nozzle flange T-101-N5 | E 18'-6-7/8"  N 26'-6"  EL 101'-0" | | |
| 10 | valve | 1"-P150-N-3015/B2 | 1"-P150-N-3015 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-N-3015/B2) on T-101-N6 | E 28'-0"  N 26'-6"  EL 123'-8-1/4" | | |
| 11 | blind | T-101-N6 | 1"-P150-N-3015 | Fit a blind between 1"-P150-N-3015/B2 and the nozzle flange T-101-N6 | E 28'-0"  N 26'-6"  EL 123'-1-7/16" | | |
| 12 | valve | 2"-P300-LS-3007/B2 | 2"-P300-LS-3007 | Close and lock VALVE, GATE, WEDGE, OS&Y 2" P300 (2"-P300-LS-3007/B2) on T-101-N7 | E 18'-5/8"  N 26'-6"  EL 102'-6" | | |
| 13 | blind | T-101-N7 | 2"-P300-LS-3007 | Fit a blind between 2"-P300-LS-3007/B2 and the nozzle flange T-101-N7 | E 18'-6-1/8"  N 26'-6"  EL 102'-6" | | |
| 14 | valve | 2"-P150-SC-3008/A2 | 2"-P150-SC-3008 | Close and lock VALVE, GATE, WEDGE, OS&Y 2" P150 (2"-P150-SC-3008/A2) on T-101-N8 | E 18'-5/8"  N 26'-6"  EL 104'-6" | | |
| 15 | blind | T-101-N8 | 2"-P150-SC-3008 | Fit a blind between 2"-P150-SC-3008/A2 and the nozzle flange T-101-N8 | E 18'-6-1/8"  N 26'-6"  EL 104'-6" | | |
| 16 | valve | 3/4"-P150-I-6032/2 | 3/4"-P150-I-6032 | Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-I-6032/2) on T-101-N9 | E 18'-1/4"  N 26'-6"  EL 107'-0" | | |
| 17 | blind | T-101-N9 | 3/4"-P150-I-6032 | Fit a blind between 3/4"-P150-I-6032/2 and the nozzle flange T-101-N9 | E 18'-7"  N 26'-6"  EL 107'-0" | | |
| 18 | valve | 3/4"-P150-I-6033/2 | 3/4"-P150-I-6033 | Close and lock VALVE, BALL, FLOATING 3/4" P150 (3/4"-P150-I-6033/2) on T-101-N10 | E 18'-1/4"  N 26'-6"  EL 117'-0" | | |
| 19 | blind | T-101-N10 | 3/4"-P150-I-6033 | Fit a blind between 3/4"-P150-I-6033/2 and the nozzle flange T-101-N10 | E 18'-7"  N 26'-6"  EL 117'-0" | | |

## Inspections

| Id | Inspection | Criteria | Basis |
|---|---|---|---|
| WO-100-01-I1 | Internal visual examination of the shell, floor, roof structure and nozzles | Findings recorded against the owner's integrity criteria | SW-NDE-4 |
| WO-100-01-I2 | Ultrasonic thickness at the 15 TMLs listed | Each reading above the owner's minimum wall | SW-WO-6 |
| WO-100-01-I3 | Floor scan for under-floor corrosion over the full floor | Owner's floor acceptance criteria | SW-WO-6 |

## Thickness monitoring locations (SW-WO-6)

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-100-01-TML01 | T-101 | Nozzle neck T-101-N1 | E 31'-7-13/16"  N 26'-6"  EL 101'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML02 | T-101 | Nozzle neck T-101-N2 | E 25'-0"  N 22'-6"  EL 123'-3-1/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML03 | T-101 | Nozzle neck T-101-N3 | E 22'-0"  N 26'-6"  EL 123'-4-5/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML04 | T-101 | Nozzle neck T-101-N4 | E 25'-0"  N 26'-6"  EL 123'-3-11/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML05 | T-101 | Nozzle neck T-101-N5 | E 18'-6-7/8"  N 26'-6"  EL 101'-0" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML06 | T-101 | Nozzle neck T-101-N6 | E 28'-0"  N 26'-6"  EL 123'-1-7/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML07 | T-101 | Nozzle neck T-101-N7 | E 18'-6-1/8"  N 26'-6"  EL 102'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML08 | T-101 | Nozzle neck T-101-N8 | E 18'-6-1/8"  N 26'-6"  EL 104'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML09 | T-101 | Nozzle neck T-101-N9 | E 18'-7"  N 26'-6"  EL 107'-0" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML10 | T-101 | Nozzle neck T-101-N10 | E 18'-7"  N 26'-6"  EL 117'-0" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-100-01-TML11 | T-101 | Floor centre | E 25'-0"  N 26'-6"  EL 100'-6" | | centre: ____ | | | |
| WO-100-01-TML12 | T-101 | Shell at 0.000" above the floor | E 25'-0"  N 26'-6"  EL 100'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-100-01-TML13 | T-101 | Shell at 6'-0" above the floor | E 25'-0"  N 26'-6"  EL 106'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-100-01-TML14 | T-101 | Shell at 12'-0" above the floor | E 25'-0"  N 26'-6"  EL 112'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-100-01-TML15 | T-101 | Shell at 18'-0" above the floor | E 25'-0"  N 26'-6"  EL 118'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |

## Test — Service leak test

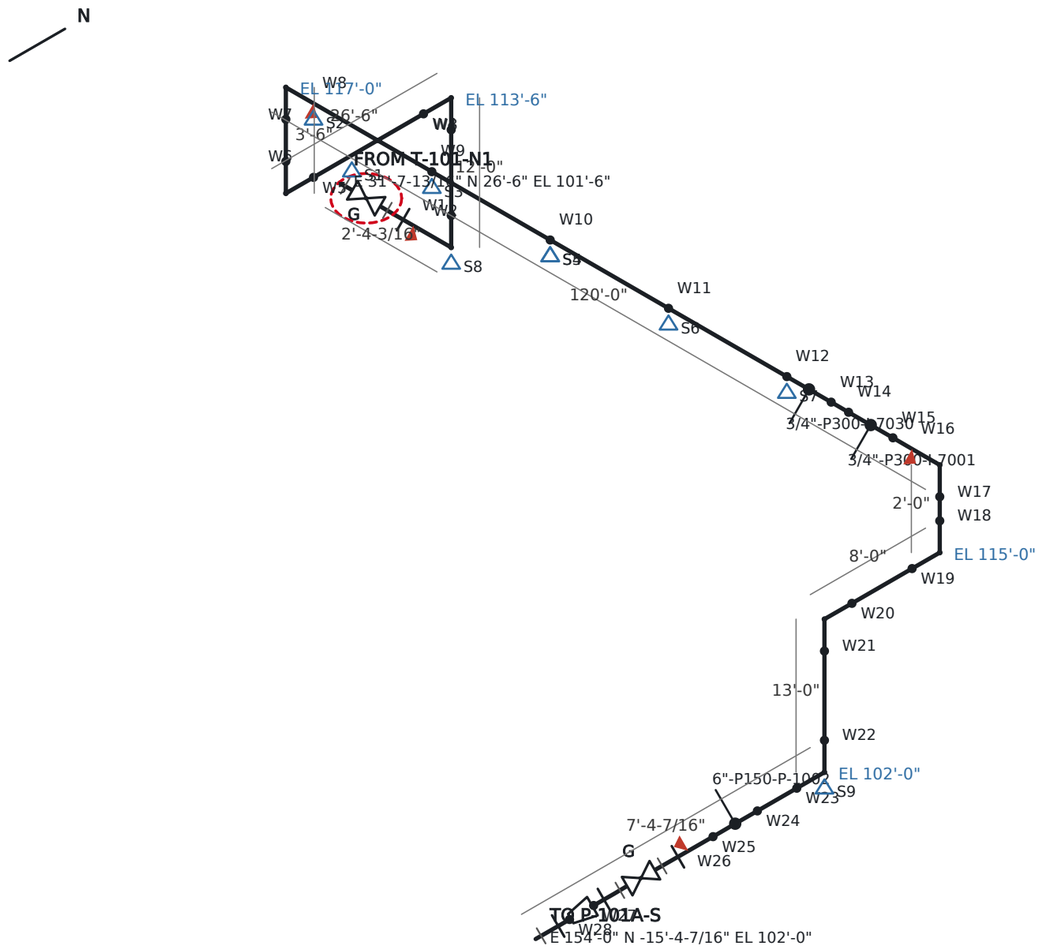
| | |
|---|---|
| **Description** | Every opened joint checked at operating pressure on start-up |
| **Basis** | SW-TEST-2 |

## Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Equipment vented, purged and gas-tested for entry; confined-space entry permit issued | Owner entry procedure | Atmosphere safe for entry per owner limits | Gas test record, entry permit | H | H | — | | | | |
| 3 | Opened for entry; cleaned for inspection | Owner procedure | Surfaces clean enough to inspect | Daily report | S | S | — | | | | |
| 4 | Internal visual examination | SW-NDE-4 | Findings recorded | Inspection report with photographs | W | H | W | | | | |
| 5 | Ultrasonic thickness at 15 TMLs | SW-WO-6 | Above the owner's minimum wall | Thickness record per TML | S | R | R | | | | |
| 6 | Floor scan | SW-WO-6 | Owner's floor criteria | Floor scan report | S | R | R | | | | |
| 7 | Owner's decision on findings (repair, monitor or accept) | Owner integrity procedure | Decision recorded | Signed disposition | R | H | R | | | | |
| 8 | Closed with new gaskets on every opened joint; studs to preload | SW-WO-5 | Preload as stated | Bolt-up record | W | W | — | | | | |
| 9 | Service leak test of opened joints on start-up | SW-TEST-2 | No leak | Leak test record | W | W | — | | | | |
| 10 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 11 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

ISOMETRIC — 6"-P150-P-1001 — Internal inspection of tank T-101




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 6" P150 COMPRESSED SHEET | 4 |
| 2 | VALVE, GATE, WEDGE, OS&Y 6" P150 | 2 |
| 3 | STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P150 | 4 |
| 4 | FLANGE WELD-NECK 6" P150 RAISED FACE | 3 |
| 5 | ELBOW 90° R = 5/3 OD 6" P150 | 8 |
| 6 | PIPE 6" P150 BEVELLED ENDS | 174'-6-1/8" |
| 7 | TEE REDUCING 6" × 3/4" P150 | 2 |
| 8 | TEE EQUAL 6" P150 | 1 |
| 9 | REDUCER CONCENTRIC 6" × 4" P150 | 1 |
| 10 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150 | 1 |
| 11 | FLANGE WELD-NECK 4" P150 RAISED FACE | 1 |
| 12 | GASKET 4" P150 COMPRESSED SHEET | 1 |

## WELDS 28 (4 FIELD) · SUPPORTS 9 · SPOOLS 6

W1F W2 W3 W4 W5 W6 W7 W8F W9 W10
W11 W12 W13 W14 W15 W16F W17 W18 W19 W20
W21 W22 W23 W24 W25 W26F W27 W28

# 6"-P150-P-1001

CLASS H150 — Hydrocarbon process
SERVICE P · 6" · P150 · DESIGN 150 psi
T-101-N1 → P-101A-S
PIPE 174'-6-1/8" · WEIGHT 3040 lb

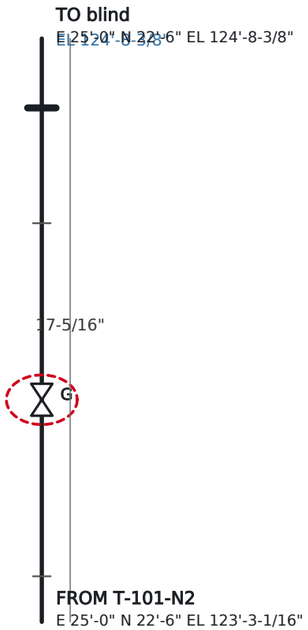
**DWG ISO-6"-P150-P-1001 REV 0 WO-100-01**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**

ISOMETRIC — 4"-P150-P-6038 — Internal inspection of tank T-101

N




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | GASKET 4" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, GATE, WEDGE, OS&Y 4" P150 | 1 |
| 3 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150 | 2 |
| 4 | FLANGE BLIND 4" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 4"-P150-P-6038

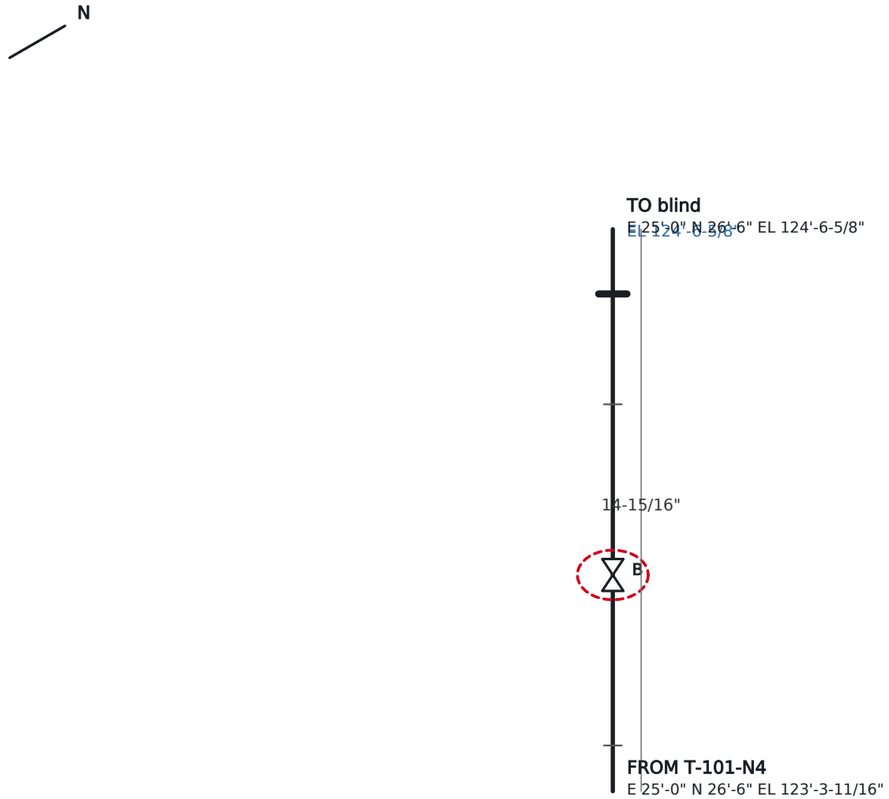
CLASS H150 — Hydrocarbon process
SERVICE P · 4" · P150 · DESIGN 150 psi
T-101-N2 → blind
PIPE 0.000" · WEIGHT 176 lb

**DWG ISO-4"-P150-P-6038 REV 0 WO-100-01**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**

ISOMETRIC — 1"-P150-V-6030 — Internal inspection of tank T-101




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | GASKET 1" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 1" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 2 |
| 4 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 1"-P150-V-6030

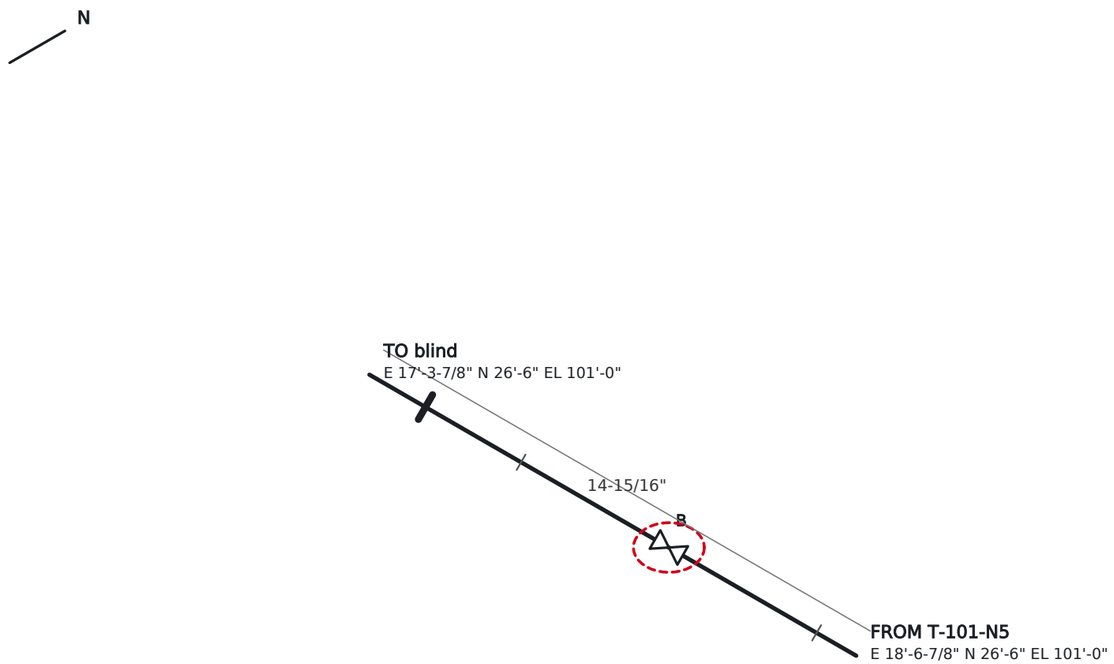
CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
T-101-N4 → blind
PIPE 0.000" · WEIGHT 39 lb

**DWG ISO-1"-P150-V-6030  REV 0  WO-100-01**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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ISOMETRIC — 1"-P150-D-6031 — Internal inspection of tank T-101




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | GASKET 1" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 1" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 2 |
| 4 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 1"-P150-D-6031

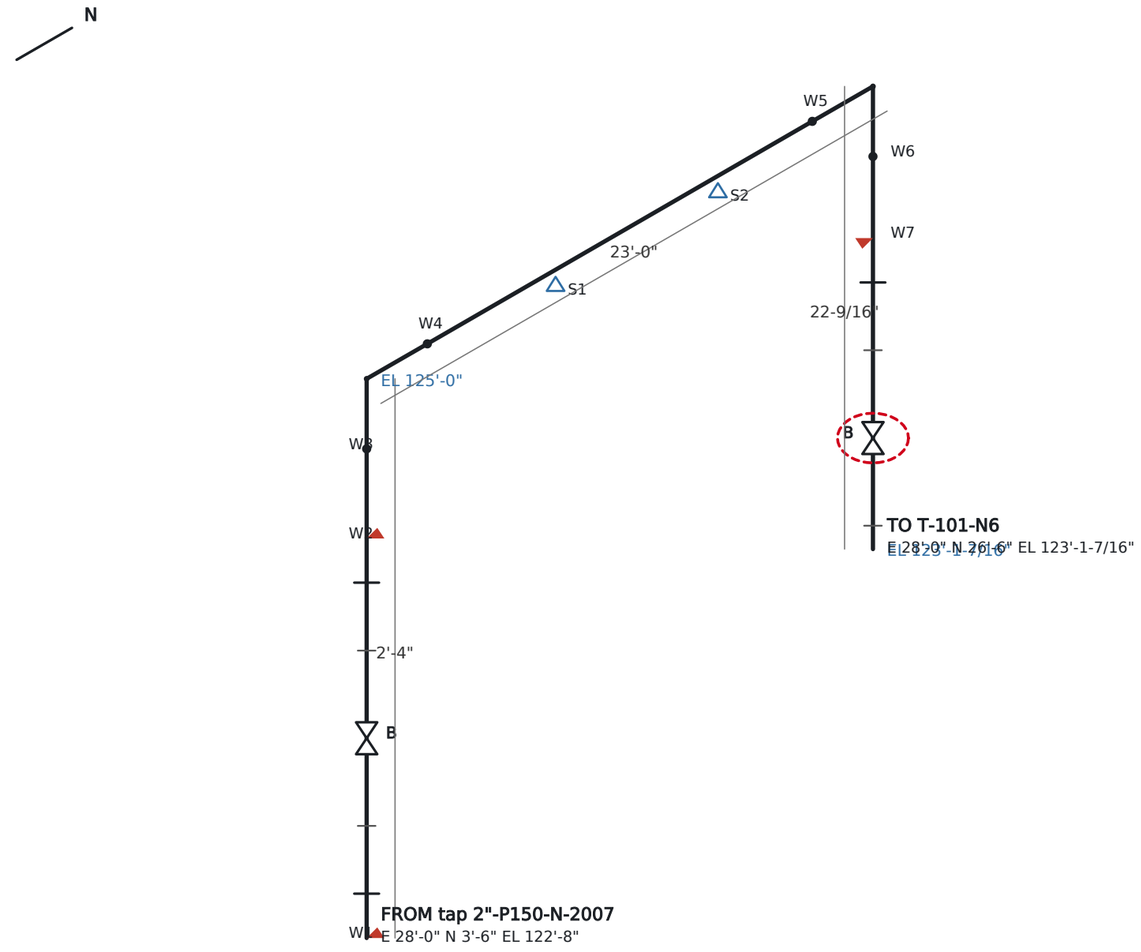
CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
T-101-N5 → blind
PIPE 0.000" · WEIGHT 39 lb

**DWG ISO-1"-P150-D-6031 REV 0 WO-100-01**

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ISOMETRIC — 1"-P150-N-3015 — Internal inspection of tank T-101




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 1" P150 RAISED FACE | 3 |
| 2 | GASKET 1" P150 COMPRESSED SHEET | 4 |
| 3 | VALVE, BALL, FLOATING 1" P150 | 2 |
| 4 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 4 |
| 5 | PIPE 1" P150 BEVELLED ENDS | 23'-4-1/8" |
| 6 | ELBOW 90° R = 5/3 OD 1" P150 | 2 |

**WELDS 7 (3 FIELD) · SUPPORTS 2 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7F

## 1"-P150-N-3015

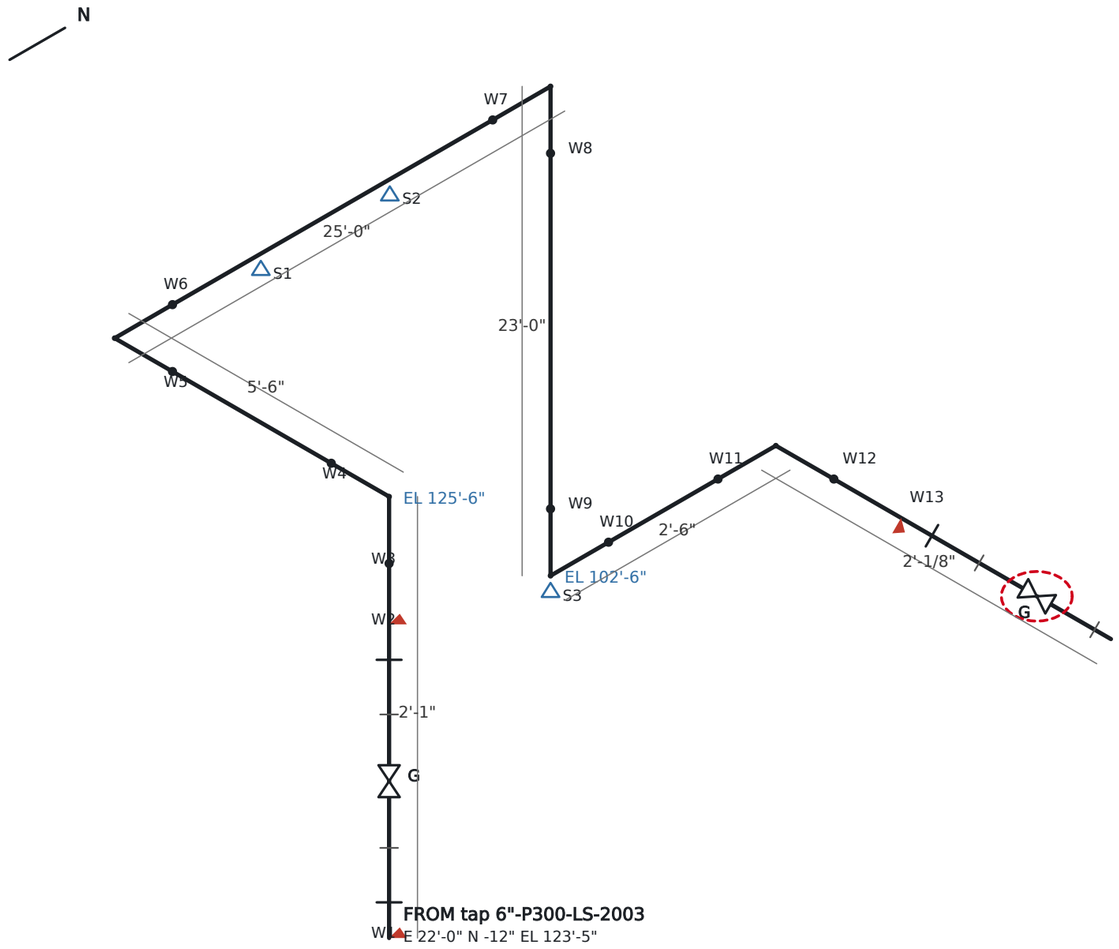
CLASS G150 — Air and gases
SERVICE N · 1" · P150 · DESIGN 150 psi
tap 2"-P150-N-2007 → T-101-N6
PIPE 23'-4-1/8" · WEIGHT 104 lb

**DWG ISO-1"-P150-N-3015 REV 0 WO-100-01**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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ISOMETRIC — 2"-P300-LS-3007 — Internal inspection of tank T-101




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 2" P300 RAISED FACE | 3 |
| 2 | GASKET 2" P300 COMPRESSED SHEET | 4 |
| 3 | VALVE, GATE, WEDGE, OS&Y 2" P300 | 2 |
| 4 | STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P300 | 4 |
| 5 | PIPE 2" P300 BEVELLED ENDS | 54'-5-1/4" |
| 6 | ELBOW 90° R = 5/3 OD 2" P300 | 5 |

**WELDS 13 (3 FIELD) · SUPPORTS 3 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F

## 2"-P300-LS-3007

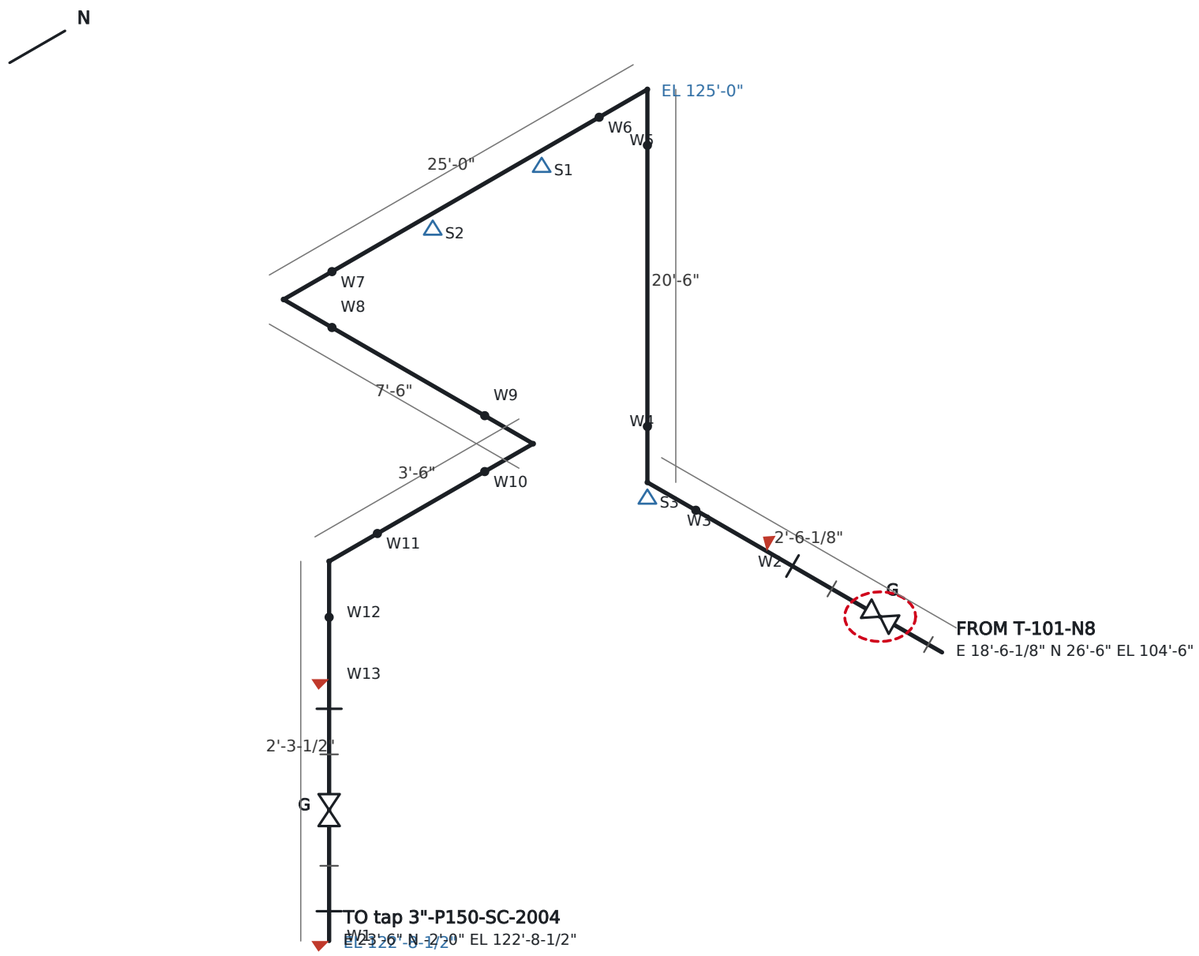
CLASS S300 — Low-pressure steam — INSULATED
SERVICE LS · 2" · P300 · DESIGN 300 psi
tap 6"-P300-LS-2003 → T-101-N7
PIPE 54'-5-1/4" · WEIGHT 297 lb

**DWG ISO-2"-P300-LS-3007 REV 0 WO-100-01**

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ISOMETRIC — 2"-P150-SC-3008 — Internal inspection of tank T-101




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | GASKET 2" P150 COMPRESSED SHEET | 4 |
| 2 | VALVE, GATE, WEDGE, OS&Y 2" P150 | 2 |
| 3 | STUD SET 4 × 0.625" × 4-1/4" WITH 2 NUTS, FOR 2" P150 | 4 |
| 4 | FLANGE WELD-NECK 2" P150 RAISED FACE | 3 |
| 5 | PIPE 2" P150 BEVELLED ENDS | 55'-7-3/4" |
| 6 | ELBOW 90° R = 5/3 OD 2" P150 | 5 |

**WELDS 13 (3 FIELD) · SUPPORTS 3 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F

## 2"-P150-SC-3008

CLASS C150 — Steam condensate — INSULATED
SERVICE SC · 2" · P150 · DESIGN 150 psi
T-101-N8 → tap 3"-P150-SC-2004
PIPE 55'-7-3/4" · WEIGHT 300 lb

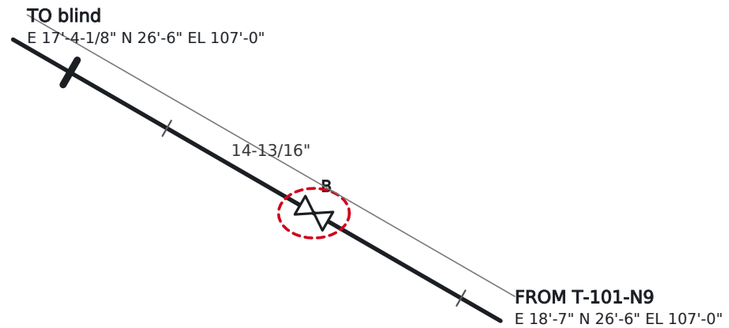
**DWG ISO-2"-P150-SC-3008 REV 0 WO-100-01**

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ISOMETRIC — 3/4"-P150-I-6032 — Internal inspection of tank T-101

N




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | GASKET 3/4" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 3/4" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150 | 2 |
| 4 | FLANGE BLIND 3/4" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 3/4"-P150-I-6032

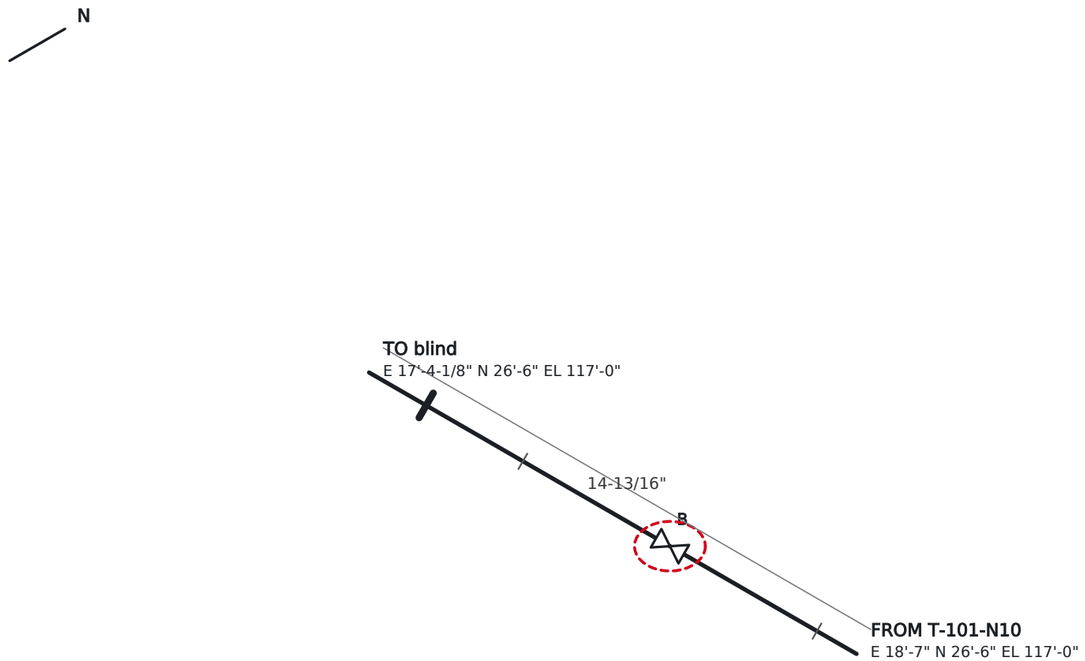
CLASS H150 — Hydrocarbon process
SERVICE I · 0.75" · P150 · DESIGN 150 psi
T-101-N9 → blind
PIPE 0.000" · WEIGHT 32 lb

**DWG ISO-3/4"-P150-I-6032 REV 0 WO-100-01**

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ISOMETRIC — 3/4"-P150-I-6033 — Internal inspection of tank T-101




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | GASKET 3/4" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 3/4" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P150 | 2 |
| 4 | FLANGE BLIND 3/4" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 3/4"-P150-I-6033

CLASS H150 — Hydrocarbon process
SERVICE I · 0.75" · P150 · DESIGN 150 psi
T-101-N10 → blind
PIPE 0.000" · WEIGHT 32 lb

**DWG ISO-3/4"-P150-I-6033 REV 0 WO-100-01**

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# Notices

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